

Applying Natural Language Programming (NLP) for Topic Modelling in a Real-life Context

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1 Introduction

This project uses the customer reviews for a gym organisation PureGym which has over 800 gyms worldwide, using two review platforms - Google (23250 reviews) and Trustpilot (16673 reviews). The objective is to exercise Topic Modelling using various Natural Language Programming (NLP) techniques, including the Large Language Models (LLMs). We have followed the exercises in this project using the order prescribed in project rubric, with some practical changes suggested during the lectures run-up to the project.

1.1 Our Approach

We have followed the list of activities given in the project's rubric, but have had to leave the LLM section to complete when memory problems were resolved. The colab notebooks shared by the course instructor have been vital in completion of our activities. We have noted down our observations in text cells throughout the project notebook.

2 Initial Data Investigation

This included, after removing 9352 rows in Google dataset with no review text, to find unique locations in both data sets (Google data set now reduced to 13898 reviews, having 512 unique locations and Trustpilot with 376 unique locations). This also included basic exercises of finding the common locations in the two data sets (310 unique common locations) and tokenization of the two data sets, finding the frequency distribution of top words used in the two data sets, using bar plot for the top few words used in PureGym reviews (Figures 1a and 1b) and generating word clouds for the two data sets (Figures 2a and 2b). This was followed by collecting the negative reviews for PureGym from these data sets and repeating the frequency distribution (Figures 3a and 3b) and word clouds (Figures 4a and 4b). The two sets of word clouds displayed a similar trend for most frequent words, but with a marked difference with the latter set highlighting some words related to negative reviews.

3 Initial Topic Modelling

Initial topic modelling used BERTopic Grootendorst (2022) on a merged data set of the negative reviews from Google and Trustpilot. In the pre-processing phase, we filtered out the reviews from the common locations and merged them to get about 3888 reviews in total. Using BERTopic for the negative reviews enabled the output of topics and their probabilities (Figures 5 - 15), bar charts visualisations of topics with five top words in each topic (Figure 16), inter-topic distance map (Figure 29a), similarity map (Figure 17) and a hierarchical clustering of related topics grouped together (Figure 18).

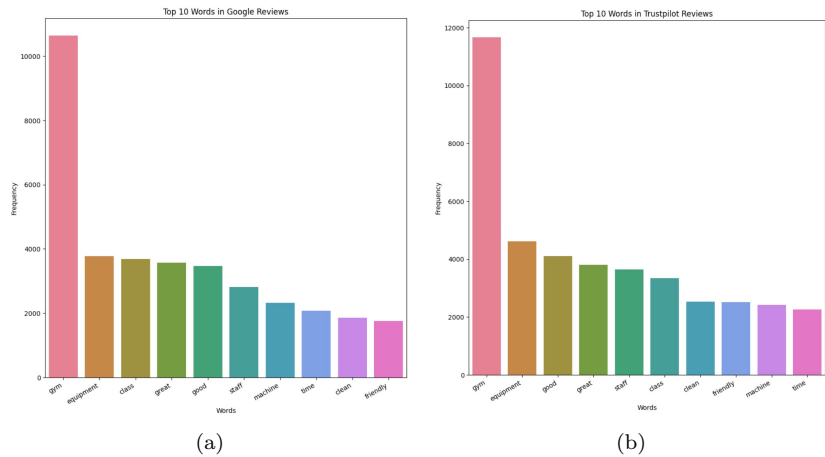


Figure 1: Bar chart for frequencies of top 10 words: (a) Google reviews (b) Trustpilot reviews

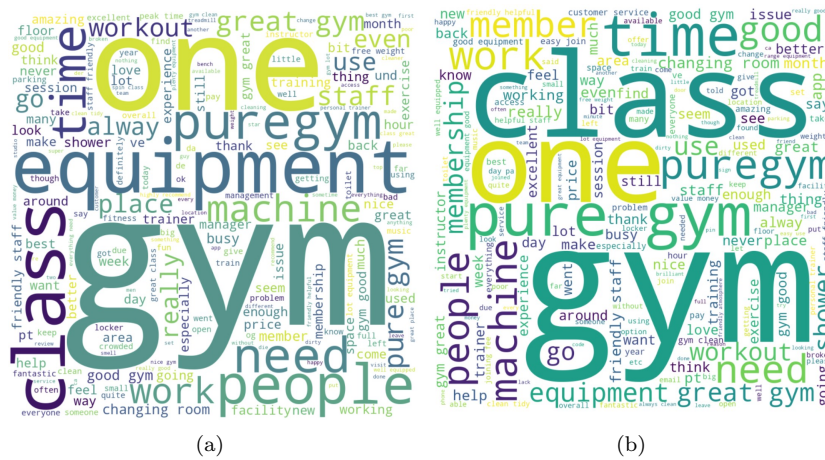


Figure 2: Wordclouds - Initial data investigation (a) Google reviews (b) Trustpilot reviews


```
[ ] model.get_topic(-1)
↔ [ ('gym', 0.017914425733818587),
    ('equipment', 0.01294043390802382),
    ('staff', 0.01146816543294195),
    ('one', 0.010676517330278622),
    ('machines', 0.010666751520563516),
    ('people', 0.010654149074799122),
    ('like', 0.009070634291397935),
    ('time', 0.008992741565009753),
    ('puregym', 0.008983982908360172),
    ('use', 0.008962926731427701)]
```

Cluster -1: The issue here seems to be people's use of machines and gym equipment which some members did not like.

Figure 5: Topic -1: Top words, their frequency distributions and possible theme.

```
[ ] model.get_topic(0)
↔ [ ('air', 0.06982350002974003),
    ('conditioning', 0.04207801946790915),
    ('hot', 0.034840833946557626),
    ('aircon', 0.03180372353855464),
    ('con', 0.027663225647846977),
    ('summer', 0.026151515171462306),
    ('ac', 0.02481041182933603),
    ('heat', 0.022923639512963428),
    ('temperature', 0.02008209781829085),
    ('working', 0.019675723828325192)]
```

Cluster 0: The theme of the topic is related to temperature settings of the gym in summer. It seems that the comment is suggesting that the gym management needs to keep a close checking on temperature setting during summer.

Figure 6: Topic 0: Top words, their frequency distributions and possible theme.

```
[ ] model.get_topic(1)
↔ [ ('class', 0.08043306196279325),
    ('classes', 0.06857033964560522),
    ('booked', 0.026029940282924376),
    ('instructors', 0.02419447287775572),
    ('instructor', 0.023119268249241643),
    ('cancelled', 0.02219983925335492),
    ('spin', 0.018568088897620367),
    ('time', 0.015302072502763848),
    ('late', 0.012446558941679727),
    ('good', 0.012296553748689647)]
```

Cluster 1: This topic is themed upon the issue that instructors cancelled the booked classes quite late in time.

Figure 7: Topic 1: Top words, their frequency distributions and possible theme.

```
[ ] model.get_topic(2)
```

```
↔ [('pass', 0.08367076415571045),  
   ('code', 0.0826246300928231),  
   ('pin', 0.07445365625896887),  
   ('day', 0.07081369993384966),  
   ('get', 0.030821192366185916),  
   ('bought', 0.028988911525747292),  
   ('access', 0.02847850189435552),  
   ('email', 0.02837562984607728),  
   ('paid', 0.02803150077078933),  
   ('help', 0.026390661487231848)]
```

Cluster 2: This issue seems to discuss the passcode for access was not emailed to them on the same day even without buying paid help.

Figure 8: Topic 2: Top words, their frequency distributions and possible theme.

```
[ ] model.get_topic(3)
```

```
↔ [('parking', 0.15393942701590407),  
   ('car', 0.056540013685845006),  
   ('park', 0.043190097892993336),  
   ('fine', 0.03681397495604163),  
   ('free', 0.03408103947024271),  
   ('fines', 0.031025628676945378),  
   ('reg', 0.029707654781804423),  
   ('ticket', 0.022628674654430456),  
   ('website', 0.020873430895184293),  
   ('enter', 0.020714363756139244)]
```

Cluster 3: The issue seems to be about car parking fines which the user had to pay by entering the ticket number and registration of the car on website.

Figure 9: Topic 3: Top words, their frequency distributions and possible theme.

```
[ ] model.get_topic(4)
```

```
↔ [('toilets', 0.06438687763075342),  
   ('toilet', 0.048657717489526034),  
   ('changing', 0.030938583842080987),  
   ('dirty', 0.027531162221346796),  
   ('rooms', 0.024940478279362875),  
   ('always', 0.023996708810019123),  
   ('cleaning', 0.02108211727115677),  
   ('smell', 0.020134217065604497),  
   ('soap', 0.01968032843107972),  
   ('roll', 0.01785546767707341)]
```

Cluster 4: The issue is related to cleaning in the toilets of the gym, it was reported that changing rooms were dirty, toilets were smelly and these needed soap and rolls regularly provided.

Figure 10: Topic 4: Top words, their frequency distributions and possible theme.


```
[ ] model.get_topic(5)
```

```
↔ [('weights', 0.046264316445878535),  
   ('bench', 0.03647369632289986),  
   ('weight', 0.03270469126929705),  
   ('equipment', 0.028636782750725755),  
   ('plates', 0.027445061338144387),  
   ('benches', 0.026944378126705328),  
   ('dumbbells', 0.02635449424395284),  
   ('enough', 0.022409062501638495),  
   ('kg', 0.021185795743595963),  
   ('racks', 0.02076602300981862)]
```

Cluster 5: This issue seemed to be about the weights equipment such as dumbbells, plates and benches were not enough or racks needed to be upgraded.

Figure 11: Topic 5: Top words, their frequency distributions and possible theme.

```
[ ] model.get_topic(6)
```

```
↔ [('music', 0.10888828107043852),  
   ('loud', 0.06471027975204309),  
   ('noise', 0.033105388459950494),  
   ('hear', 0.028601964687346413),  
   ('volume', 0.02561105237329543),  
   ('classes', 0.025406343516078567),  
   ('headphones', 0.024188216130334575),  
   ('class', 0.01956203674678453),  
   ('gym', 0.01794639840690059),  
   ('play', 0.016724901950779043)]
```

Cluster 6: This issue refers to the level of noise and volume of music in the gym. One had to use headphones in the gym.

Figure 12: Topic 6: Top words, their frequency distributions and possible theme.

```
[ ] model.get_topic(7)
```

```
↔ [('closed', 0.11239315006423685),  
   ('open', 0.03782412028109491),  
   ('days', 0.03718433724992407),  
   ('christmas', 0.03667969178008038),  
   ('day', 0.028822118918401498),  
   ('gym', 0.02841950584710509),  
   ('opening', 0.027154203667385808),  
   ('email', 0.023071907270295483),  
   ('closing', 0.01976325915383056),  
   ('th', 0.019696652109817882)]
```

Cluster 7: There was no email about the opening/closing days by the gym during Christmas period.

Figure 13: Topic 7: Top words, their frequency distributions and possible theme.

```
[ ] model.get_topic(8)
↔ [('manager', 0.030793680002510312),
    ('rude', 0.03066444102603736),
    ('member', 0.02190523633512697),
    ('gym', 0.02059849246951974),
    ('staff', 0.01837660228527414),
    ('said', 0.017274670992155107),
    ('attitude', 0.016476451869345807),
    ('personal', 0.015306786025699782),
    ('guy', 0.015023013010115308),
    ('made', 0.013971700843804094)]
```

Cluster 8: This review mentioned that the member of staff was rude and made personal remarks. This was reported to the manager.

Figure 14: Topic 8: Top words, their frequency distributions and possible theme.

```
[ ] model.get_topic(9)
↔ [('smell', 0.06695291869354676),
    ('smells', 0.044408670361938574),
    ('ventilation', 0.03502087535624316),
    ('stinks', 0.029644936989001745),
    ('gym', 0.026870045729995694),
    ('air', 0.025618567300985683),
    ('bad', 0.0242132272437755),
    ('smelly', 0.02204399949647303),
    ('this', 0.019382232863400014),
    ('ever', 0.01754702289988443)]
```

Cluster 9: The issue here was about the air quality in the gym and the smell in the gym needed ventilation as the smell attacks in that gym.

Figure 15: Topic 9: Top words, their frequency distributions and possible theme.

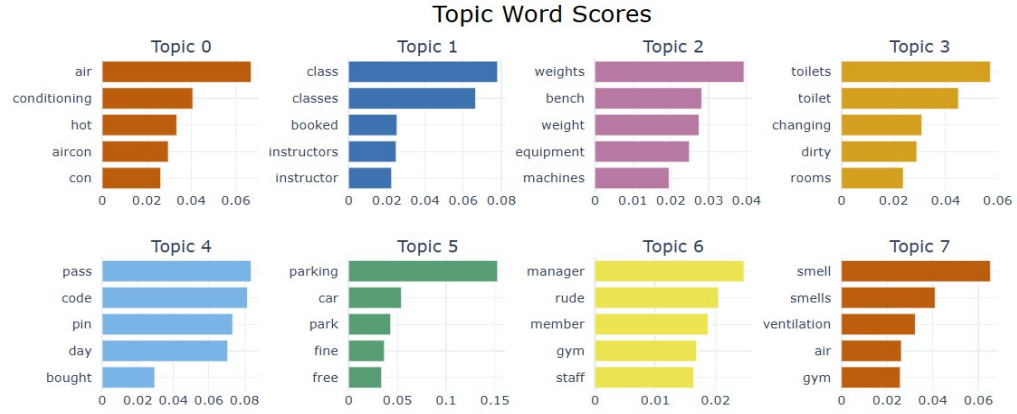


Figure 16: Top 5 five words in top 10 topics.

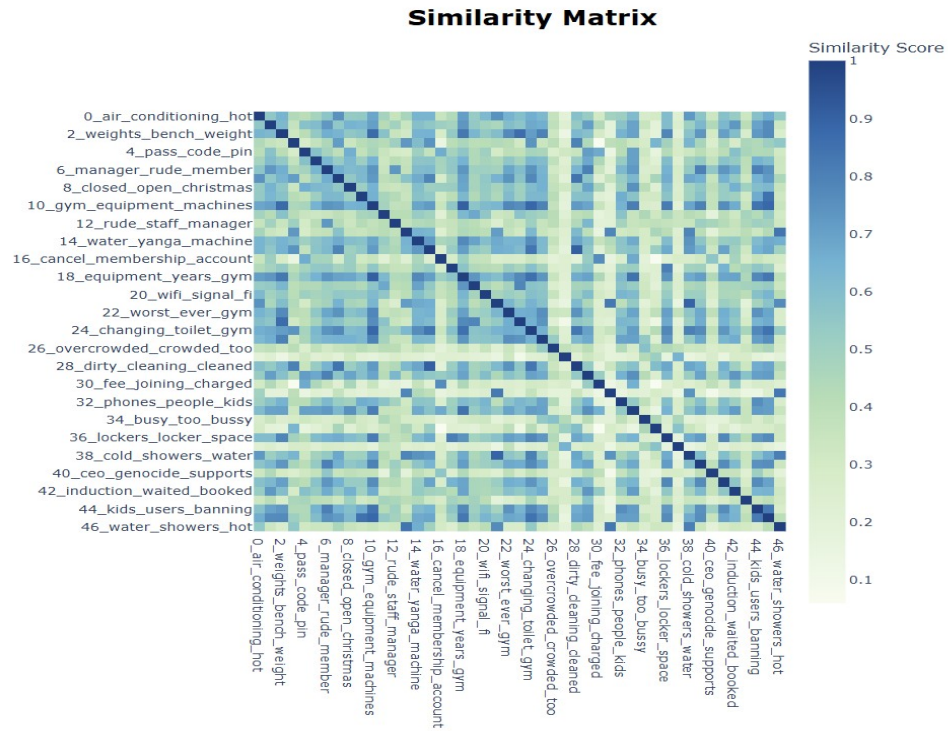


Figure 17: Similarity matrix for topics

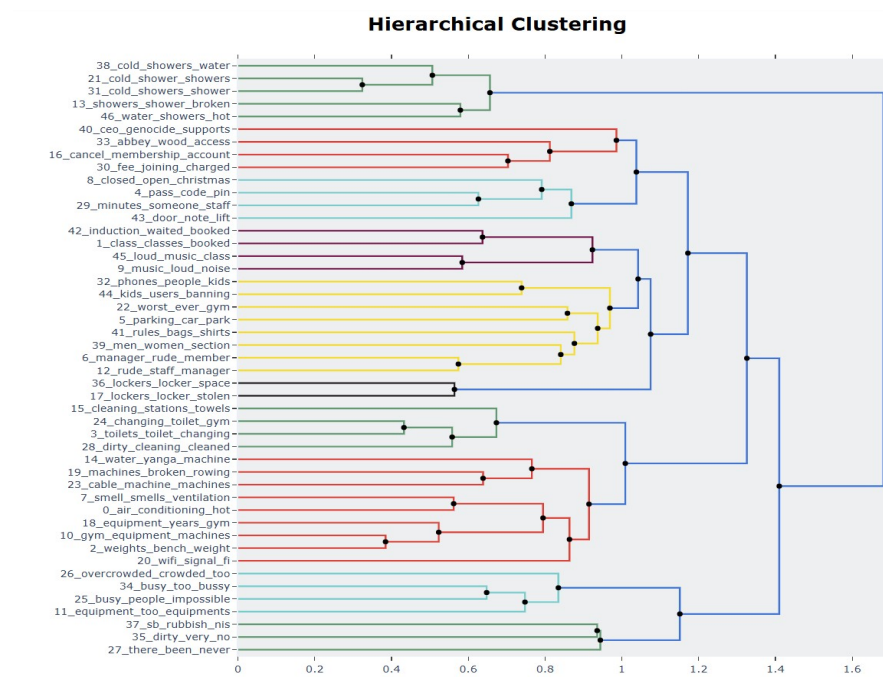


Figure 18: Hierarchical clustering of topics

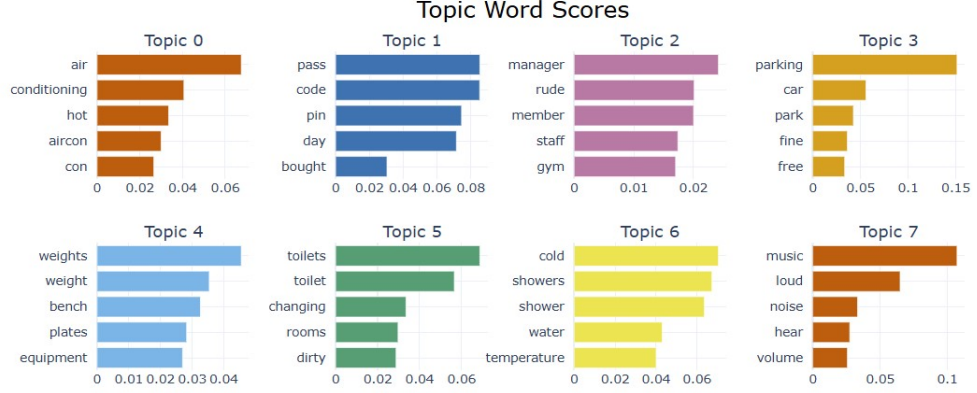


Figure 19: Top 5 five words in top 10 topics in top 30 locations of negative reviews.

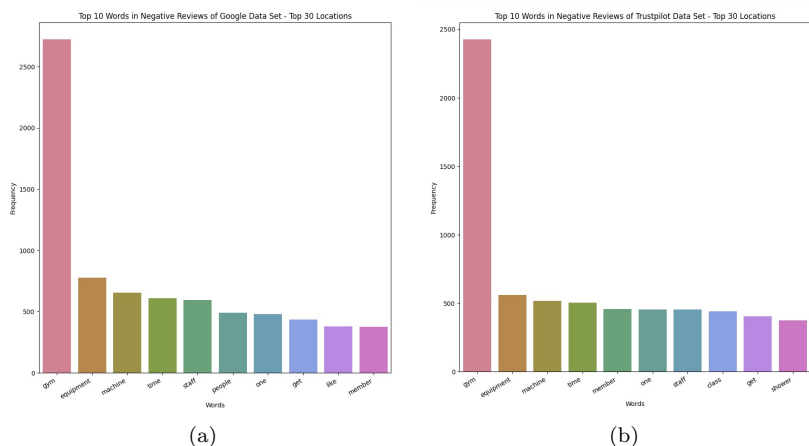
4 Further Data Investigation

In this section, top 20 locations with the highest number of reviews was listed out for both Google and Trustpilot reviews. The two datasets were merged using the Club's Name and Location Name columns of Google and Trustpilot respectively. For every location, number of Google and Trustpilot reviews were counted and summed together. This list was sorted on the total number of reviews.

In the second part of this section, the word frequency and word cloud for top 30 locations for Google and Trustpilot were displayed again. Moreover, for the top 30 locations, the reviews of Google and Trustpilot were collected together, resulting in 3888 negative reviews, and BERTopic was run on these reviews. Similar topics topped the lists, which is depicted through the bar charts of 10 topics showing top 5 words in each topic (Figure 19). However, the intertopic distance had changed (Figure 29b). The bar charts for top 10 words in Top 30 locations are depicted for Google (Figure 20a) and Trustpilot (Figure 20b). The word clouds for Google (Figure 21a) and Trustpilot (Figure 21b) showed small variation as compared to the overall corresponding reviews.

5 Conducting Emotion Analysis

The BERT model `bhadresh-savani/bert-base-uncased-emotion` from Hugging Face was used for both data sets to capture the top emotion for each review. We were able to depict bar charts of top emotions from the negative reviews of both Google (Figure 22) and Trustpilot (Figure 23) data sets.



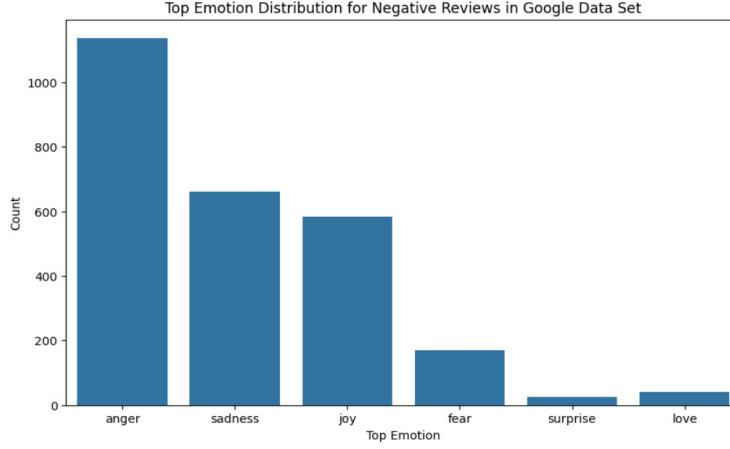


Figure 22: Top emotion frequencies shown in a bar chart for negative reviews of Google data set.

We selected those reviews from both data sets with ‘anger’ as top emotion and merged those lists. Running the BERTopic model on this merged list produced a variation in results which is depicted in the intertopic distance map (Figure 29c), top 5 words with in first 8 topics (Figure 24) and hierarchical clustering (Figure 25).

6 Using a Large Language Model (LLM) from Hugging Face

We used Phi3.5-mini-instruct model, Abdin et al. (2024), instead of falcon-7b-instruct, initially without Google Pro but without success after running for 6 hours. As this is a real-world project with over 6000 reviews, we opted for running the notebook with Google Pro’s setting for A100 GPU with 40GB memory. We resorted to running the model for a subset of 1000 reviews. The model took 35 minutes and was successful in collecting 950 topics out of 1000 as some of the reviews were out of the limit of 1000 characters that we had set on the review text.

Running the BERTopic model on the topic strings resulted above from the Phi3.5-mini-instruct model presented some variation in the top 8 topics. Two of the topics are shown in Figure 26a and Figure 26b. The hierarchical clustering result was too large to be displayed in this report, and can be viewed in the main notebook.

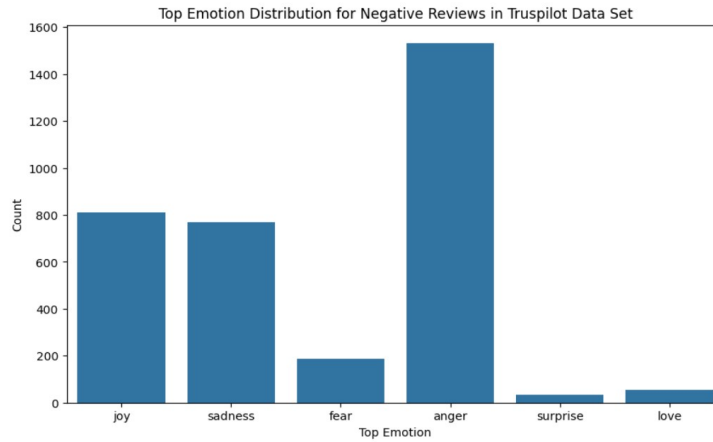


Figure 23: Top emotion frequencies shown in a bar chart for negative reviews of Trustpilot data set.

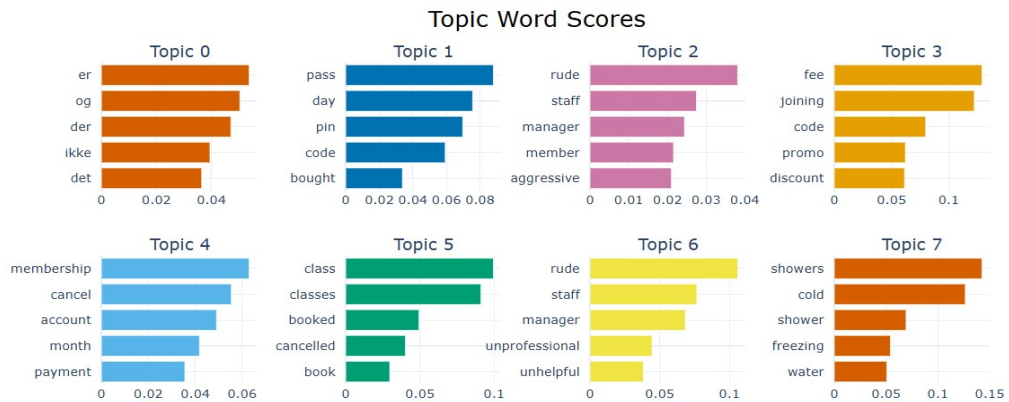


Figure 24: Top 5 five words in top 10 topics of negative reviews with anger as top emotion.

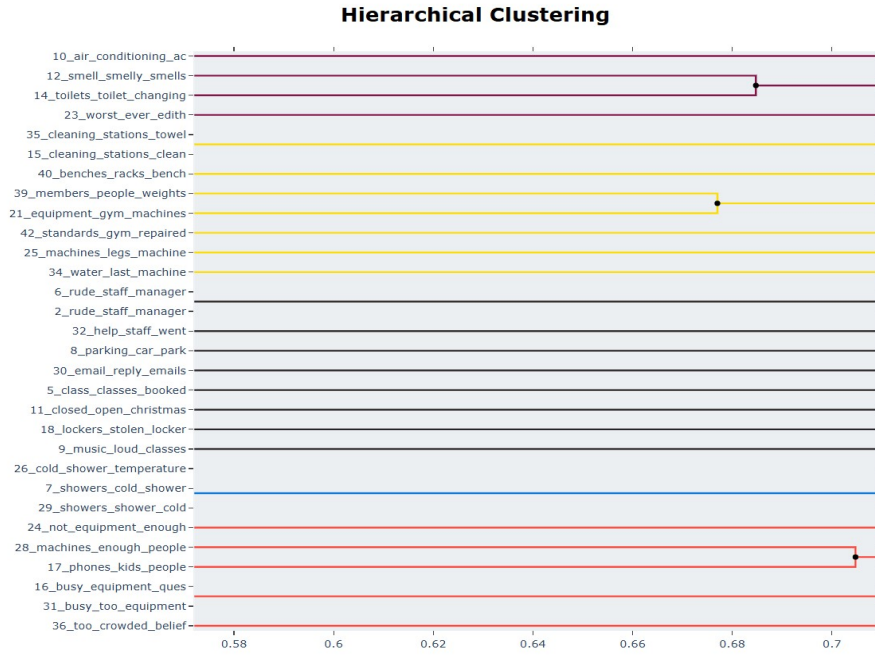


Figure 25: Hierarchical clustering of topics in Emotions Analysis - negative reviews with anger as top emotion.

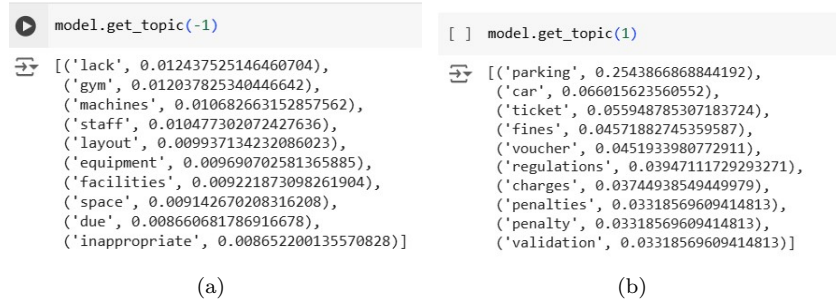


Figure 26: BERTopic words generated from the Phi3.5-mini-instruct large language model output.

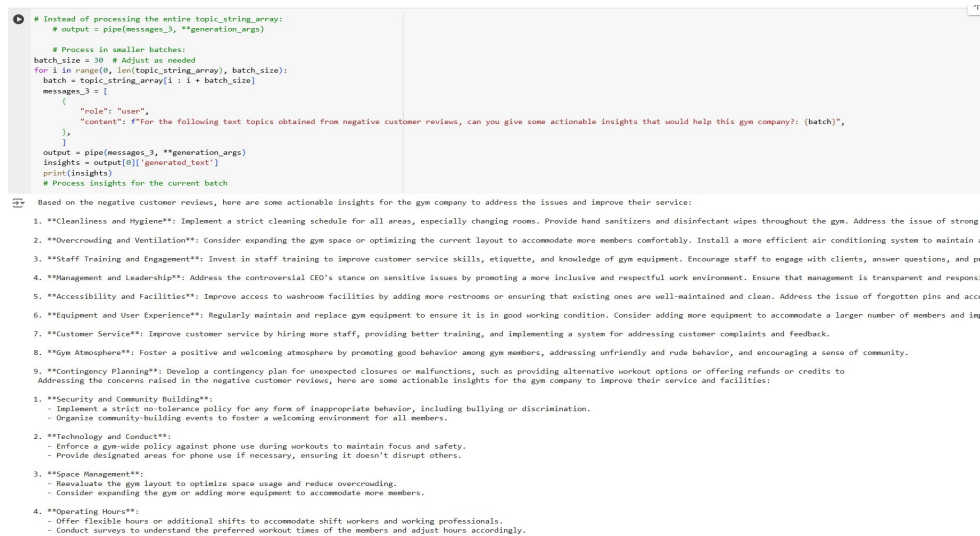


Figure 27: Batchwise insights creation by the Phi3.5-mini-instruct LLM to avoid memory error.

With the ‘out of memory’ error while attempting to derive actionable insights, we resorted to batch-wise creation and display of actionable insights, with a batch size of 30. A few are shown in Figure 27.

7 Using Gensim

In order to run the Gensim LDA model Blei (2012), we preprocessed the model to collect all the negative reviews in one list. Figure 28 provided an interactive intertopic distance map with an accompanying list of words highlighted for the selected topic.

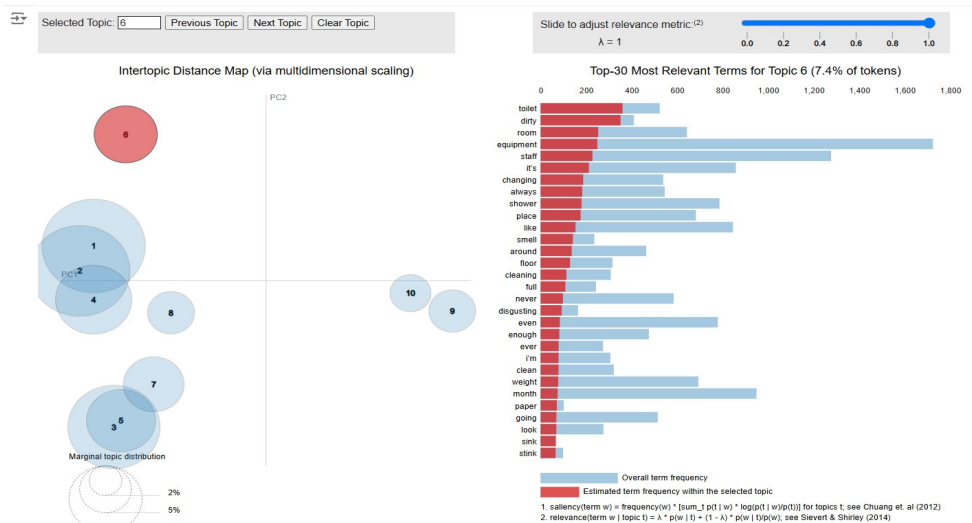


Figure 28: LDA applied for the negative reviews for cobined data set using Gensim.

8 Conclusions

Various techniques were tried in this project to analyse PureGym reviews, with the specific focus on negative reviews. BERTopic was run on various subsets of reviews to identify differences in outputs in the form of word clouds, bar charts for top 5 words in various topics and intertopic distance maps (ITDMs). The same outputs were obtained after running BERTopic on the topic strings generated by the Phi3.5-mini-instruct large language model (Figure 29d). The ITDM by the LLM showed less distance among words in negative reviews in various topics, this meant that there were more topics containing different words to describe the features that invited actionable insights. Other contained ITDMs provide only a topic analysis.

The frequency bar charts of top emotions generated for the negative reviews by Google and Trustpilot included an interesting ‘joy’ emotion, which can probably be explained. We conclude that some people who gave an overall negative score (less than 3) to the gym and its facilities, they may have written some positive comments for the gym before writing their issues and concerns.

We would like to run more large language models such as falcon-7b-instruct and other models on these data sets and compare our results with Phi3.5-mini-instruct. This is a real-life project that provides an excellent opportunity to machine learning students to experiment upon it using other models.

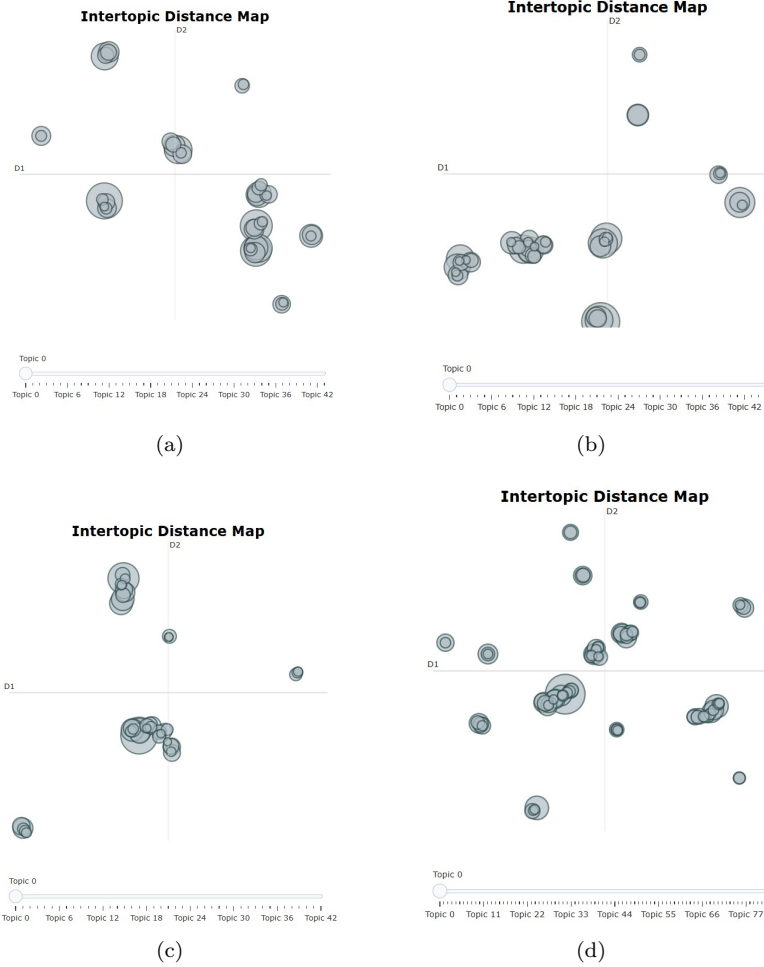


Figure 29: Intertopic distance maps generated by BERTopic: a) for all reviews (Section 2); b) for negative reviews (Section 3); c) for emotions analysis (Section 5); d) for topics strings generated by the Phi3.5-mini-instruct LLM (Section 6).

References

- Abdin, M., Aneja, J., Awadalla, H., Awadallah, A., Awan, A. A., Bach, N., Bahree, A., Bakhtiari, A., Bao, J., Behl, H., Benhaim, A., Bilenko, M., Bjorck, J., Bubeck, S., Cai, M., Cai, Q., Chaudhary, V., Chen, D., Chen, D., Chen, W., Chen, Y.-C., Chen, Y.-L., Cheng, H., Chopra, P., Dai, X., Dixon, M., Eldan, R., Frago, V., Gao, J., Gao, M., Gao, M., Garg, A., Giorno, A. D., Goswami, A., Gunasekar, S., Haider, E., Hao, J., Hewett, R. J., Hu, W., Huynh, J., Iter, D., Jacobs, S. A., Javaheripi, M., Jin, X., Karampatziakis, N., Kauffmann, P., Khademi, M., Kim, D., Kim, Y. J., Kurilenko, L., Lee, J. R., Lee, Y. T., Li, Y., Li, Y., Liang, C., Liden, L., Lin, X., Lin, Z., Liu, C., Liu, L., Liu, M., Liu, W., Liu, X., Luo, C., Madan, P., Mahmoudzadeh, A., Majercak, D., Mazzola, M., Mendes, C. C. T., Mitra, A., Modi, H., Nguyen, A., Norick, B., Patra, B., Perez-Becker, D., Portet, T., Pryzant, R., Qin, H., Radmilac, M., Ren, L., de Rosa, G., Rosset, C., Roy, S., Ruwase, O., Saarikivi, O., Saied, A., Salim, A., Santacroce, M., Shah, S., Shang, N., Sharma, H., Shen, Y., Shukla, S., Song, X., Tanaka, M., Tupini, A., Vaddamanu, P., Wang, C., Wang, G., Wang, L., Wang, S., Wang, X., Wang, Y., Ward, R., Wen, W., Witte, P., Wu, H., Wu, X., Wyatt, M., Xiao, B., Xu, C., Xu, J., Xu, W., Xue, J., Yadav, S., Yang, F., Yang, J., Yang, Y., Yang, Z., Yu, D., Yuan, L., Zhang, C., Zhang, C., Zhang, J., Zhang, L. L., Zhang, Y., Zhang, Y., Zhang, Y. & Zhou, X. (2024), ‘Phi-3 technical report: A highly capable language model locally on your phone’.
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- Blei, D. M. (2012), ‘Probabilistic topic models’, *Communications of the ACM* **55**(4), 77–84.
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